

From Particles to Waves

Towards Coupling Dynamic
Snow Modelling and Spectral
Elements to Detect
Avalanche Breakoff in DAS
Data

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Why Model Crack Propagation in Snow? - Relevance

- 22 fatalities by avalanche in Switzerland every year (**LongtermStatistics**)
- Avalanches are danger for human life but also infrastructure
(**schweizerSnowAvalancheFormation2003**)
- Understanding processes associated with avalanches important



What is Crack Propagation in Snow? - Relevance

- Avalanche release because of loading, leading to crack propagation in a weak layer (**gaumeSnowFractureRelation2017a**)
- Crack propagation at different speeds: sub-Rayleigh (below vs of material) and Supershear (above vs) (**trottetTransitionSubRayleighAnticrack2022**), transition between two regimes possible (**bergfeldSignaturesSubRayleighSupershear2025**)
- Fast crack propagation → critical crack length for release (**gaumeModelingCrackPropagation2015**) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (**paitzPhenomenologyAvalancheRecordings2023**; **edmeFiberopticDetectionSnow2023**) and Norway (**turquetAutomatedSnowAvalanche2024**)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

Overview

- **Goal:** Couple dynamic snow modelling with SE to model crack propagation in snow
- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations

Nomenclature & Abbreviations

- **SE, FE, FEM**: Spectral elements, Finite Element (Method), done in SALVUS
- **MPM**: Material Point Method Numerical Modeling
- **PST**: Propagation Saw Test
- **F-k**: Frequency-Wavenumber Plot
- **DAS**: Distributed Acoustic Sensing (Fibre-Optics)
- **STF**: Source-time function (the wavelet fired by SALVUS)

Research Questions

- What FE modelling framework is required to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
- What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?
- To what extent do the results change if source wavelets from a MPM simulation of a PST are injected?
- How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

Simple Model I

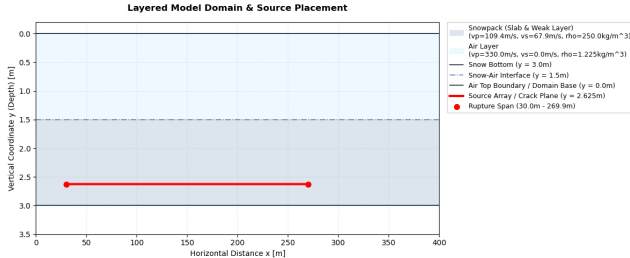


Figure: Two-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Moving source line extent in red. Propagation speeds according to **trottetTransitionSubRayleighAnticrack2022**.

- SALVUS Output:
Displacement field u , used to simulate DAS using Pyber library by **noeFrontiersFullwaveformSeismology**
- $\frac{\partial u_i}{\partial t} \rightarrow$ velocity in x and y
- $\frac{\partial u_i}{\partial j} \rightarrow$ strain in x and y (ϵ)
- $\frac{\partial^2 u_i}{\partial t \partial j} \rightarrow$ strain rate ($\dot{\epsilon}$)

Simple Sub-Rayleigh: Strain

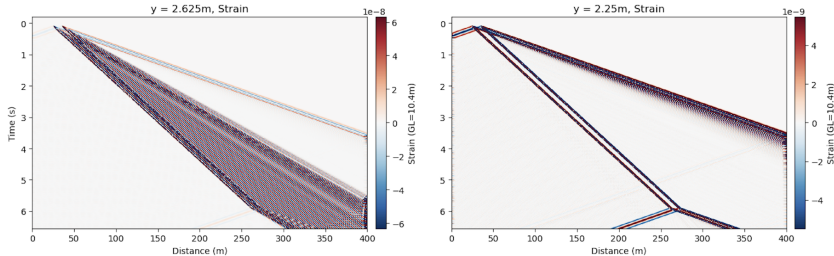
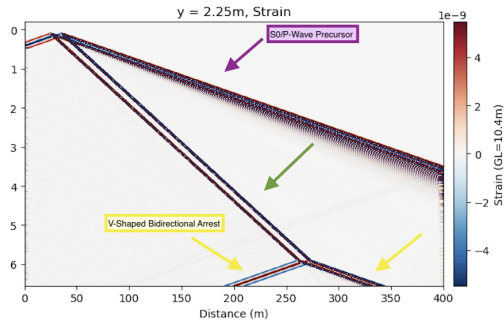
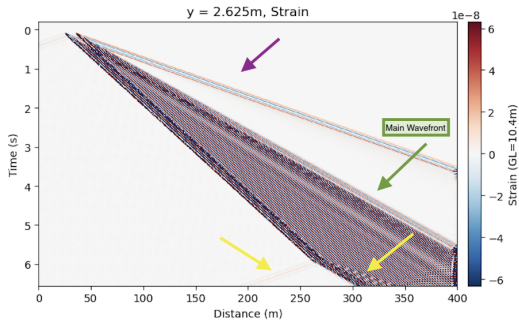


Figure: Horizontal Strain for sub-Rayleigh crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.



- Precursor emitted by all sources but they interfere (**madariagaDynamicsExpandingCircular1976**), as shown by wider source spacing
- Slope of line close to p-wave speed in medium: S0 wave bounded by plate wave speed limit (**roseWavesPlates2014**)
- S0/P-wave precursor observed in field data by **herwijnenSeismicSensorArray2011**
- Significant weakening of amplitude between two depths

Simple model: Lamb Wave Identification I

- Polarity oscillation in all plots: indicative of Lamb waves
- Plotting dispersion curves over f-k spectra: energy is concentrated at symmetric (S0) and antisymmetric (A0) mode
- Antisymmetric mode gets excited preferentially when source is applied normal to midplane of material (**lambWavesElasticPlate1917**; **roseWavesPlates2014**; **achenbachWavePropagationElastic2012**), so in this case yy
- Confirmed by animations of wavefield:

Simple model: Lamb Wave Identification II

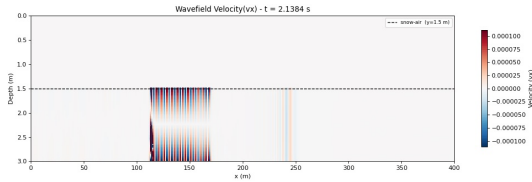


Figure: Animation still for Sub-Rayleigh Vx

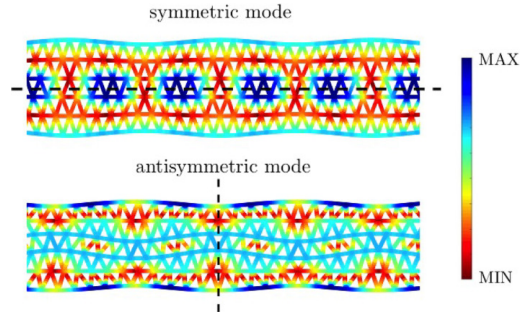


Figure: Lamb wave modes intensity, taken from **cartaLambWavesDiscrete2023**

Simple Supershear: Strain

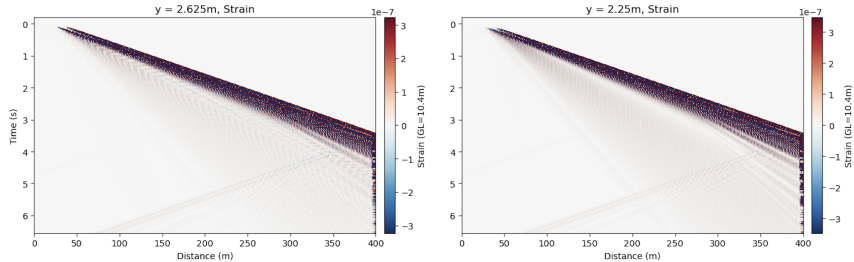
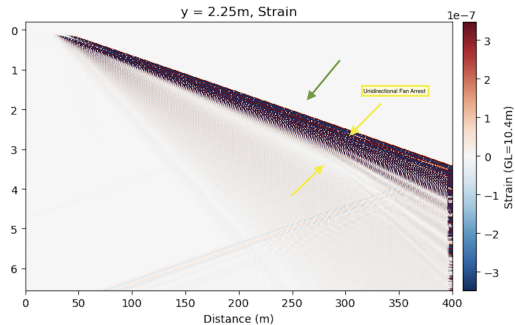
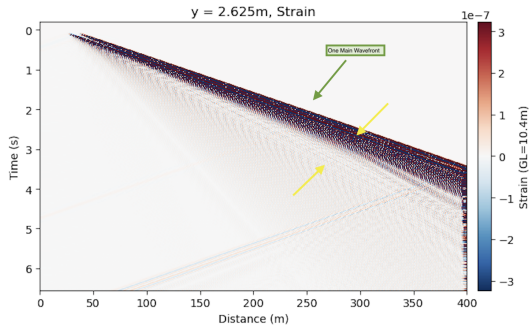


Figure: Horizontal Strain for Supershear crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.



- Weakening and broadening of main wavefront (**kAkiRichards19801981**)
- At $x=270\text{m}$, unidirectional fan-like arrest front: rupture outruns vs, so arrest energy is only propagated forward (**madariagaHighfrequencyRadiationCrack1977**)
- Supershear larger amplitude compared to sub-Rayleigh

MPM-Guided Model I

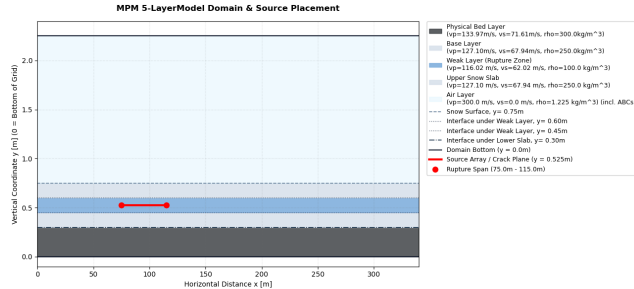


Figure: Five-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

- Source-time functions obtained from MPM-simulation of PST by **trottetTransitionSubRayleighAnticra**
- Source Wavelet in all three directions, but irregular and different for every source point

MPM-Guided: Strain

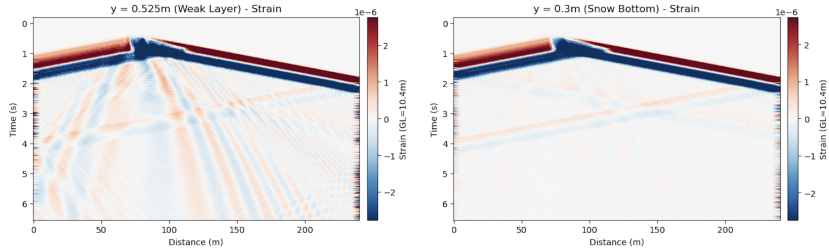
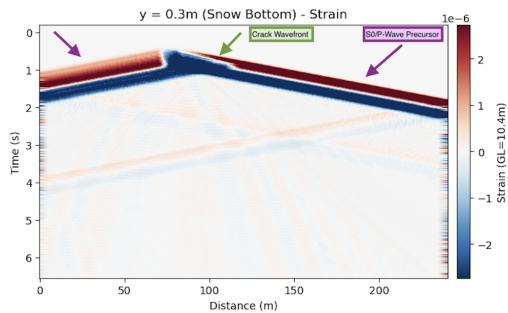
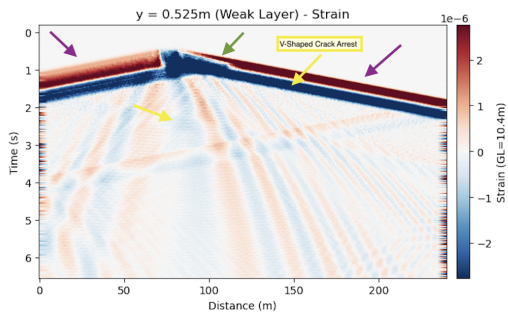


Figure: Horizontal Strain for MPM-guided model. Plotted at crack depth (0.525) and at the snow bottom (0.3m). Gauge length is 10.4m.



- Static field = permanent deformation left after the source stops radiating
(**zhuNoteDynamicStatic2002**; **archambeauGeneralTheoryElastodynamic1968**)
- Ricker wavelets (simple model) are symmetric $\rightarrow$ zero net moment $\rightarrow$ moment returns to 0 $\rightarrow$ no static field
- MPM-derived moment components are asymmetric, no return to 0 $\rightarrow$ nonzero net (DC) moment $\rightarrow$ permanent displacement (**zhuNoteDynamicStatic2002**; **kAkiRichards19801981**)
- All three MPM moment components settle at a negative value rather than returning to zero: net moment $\rightarrow$ weak-layer collapse, consistent with **heierliAnticracksNewTheory2003**; **heierliAnticrackNucleationTriggering2008**; **gaumeDynamicAnticrackPropagation2018**; **trottetTransitionSubRayleighAnticrack2022**
- All components share this same sign (no alternation, as mode II shear would show) $\rightarrow$ mixed-mode anticrack, not pure opening or shearing (mode I/II)

MPM-Guided Model: Lamb Wave Identification

- Like in simple model: oscillatory wavefronts $\rightarrow$ A0 mode

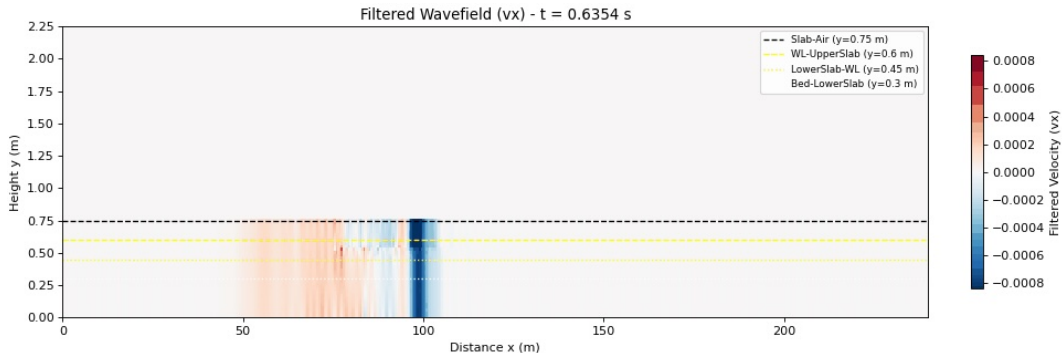
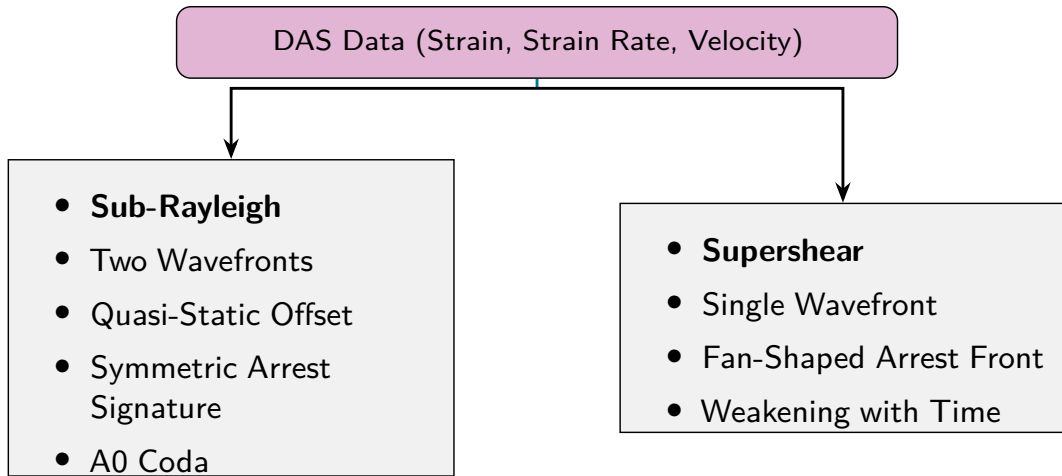


Figure: Animation still for MPM V_x

Summary of Results & Discussion I

- S0 Precursor observed in all Strain & Strain Rate Plots, as well as in traces
- Main wavfront in all cases is A0 $\rightarrow$ source normal to slab midplane
- Different rupture arrest structures depending on propagation speed
- MPM introduces plastic and static deformation
- Weakening of amplitude only in simple case $\rightarrow$ also thicker slab

Implications for DAS field data



Outlook

- Add topography to models
- Verify Supershear effects for more complex STFs as well
- Inhomogeneous, multi-layered snowpack
- Transitional propagation regimes
- Comparison to field data → amplitude of signals compared to noise
- 3D model

Answer to Research Questions - Conclusions

- **What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?** Rupture-arrest signatures depending on propagation speed, S0 precursor, A0 main wavefront for both propagation regimes. Changes in amplitude
- **To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?** Verification of sub-Rayleigh S0 precursor, A0 mode excitation, changes because appearance of quasi-static offset
- **How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?** Precursors (S0 and quasi-static offset) used as "early" warning, identification of propagation speed using arrest structures, possibility of identifying transitions by number of wavefronts

Questions?

Answer to Research Questions - Conclusions

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Bibliography I

MPM STF Workflow

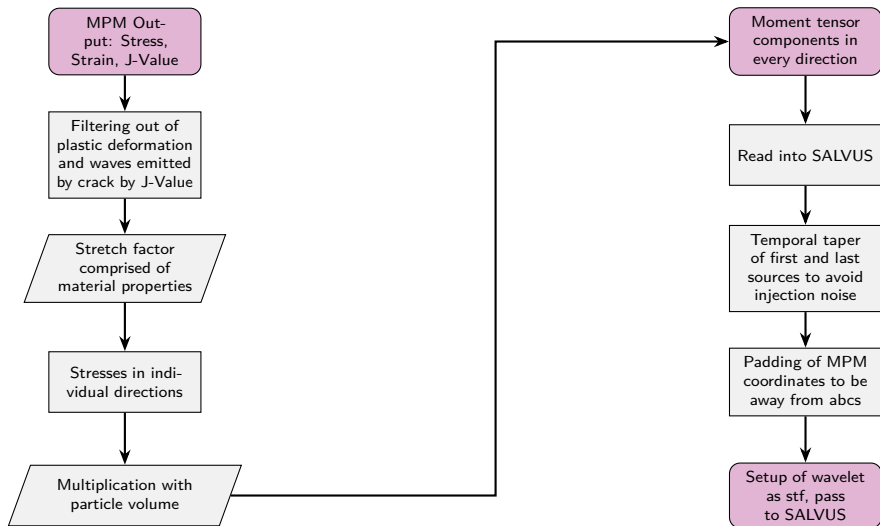
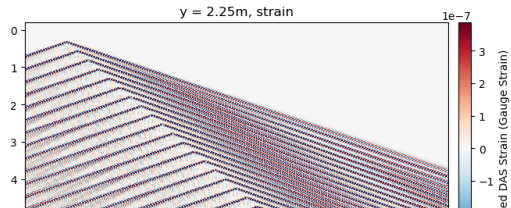
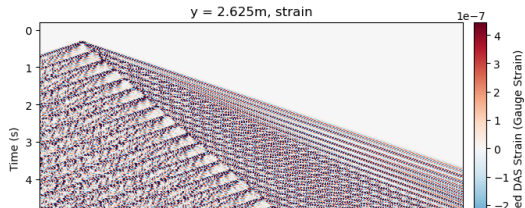
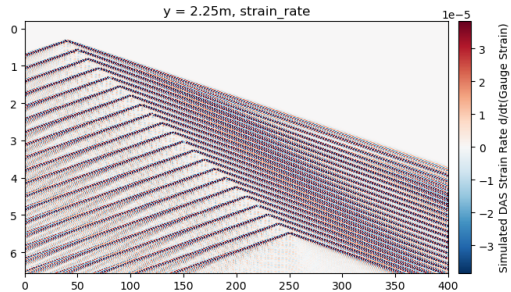
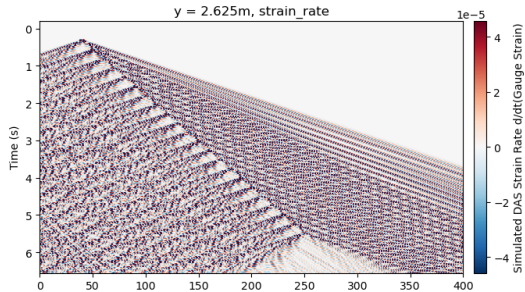


Figure: Workflow for conversion of MPM data and input as SALVUS STF.

Wider Source Spacing

**Sub-Rayleigh: Pyber DAS Simulation (Gaussian, width=4.0m)
at $y = 2.625\text{m}$ (Crack Depth) and $y = 2.25\text{m}$ (Snow Bottom)**



Simple Supershear Trace

Supershear: Traces at First 5 Receivers (70.0m - 71.0m)

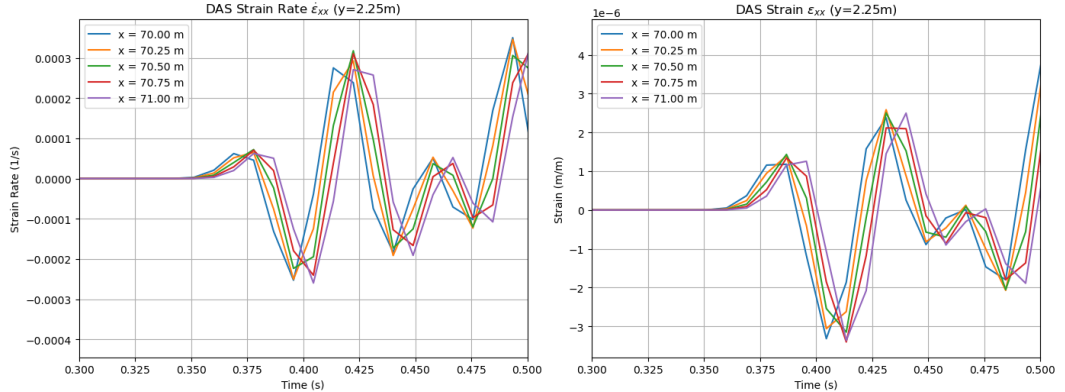
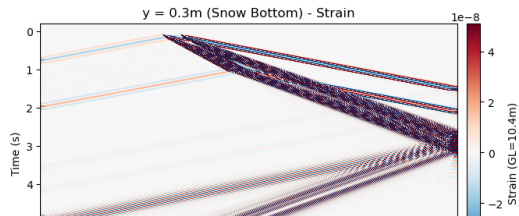
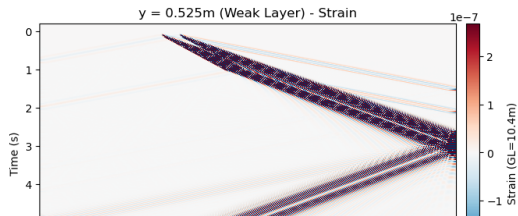
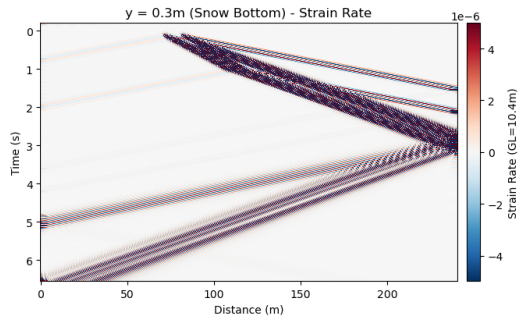
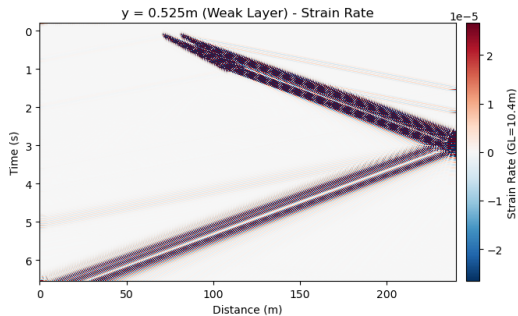


Figure: Strain and Strain rate trace for first 5 receivers in Supershear case

Simple Source in 5 Layer Geometry

5-Layer Model: Simulated DAS Strain Rate and Strain at $y = 0.525\text{m}$ and $y = 0.3\text{m}$



Simple model velocity plots I

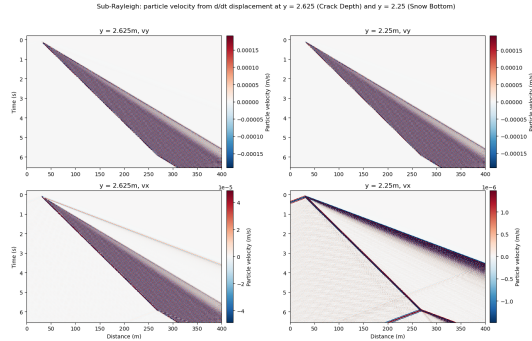


Figure: Simple model Sub-Rayleigh velocity plots. Crack depth on left side, snow bottom on right. v_y at top, v_x at bottom. Velocity derived from displacement.

Simple model velocity plots II

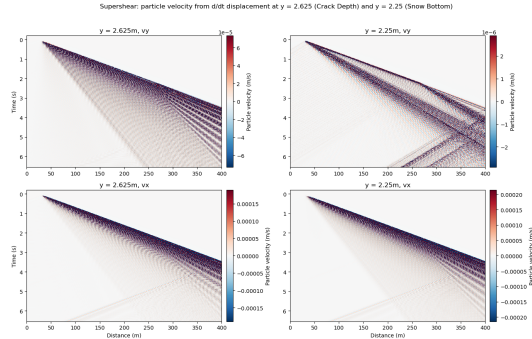


Figure: Simple model Supershear velocity plots. Crack depth on left side, snow bottom on right. v_y at top, v_x at bottom. Velocity derived from displacement.

MPM velocity plots

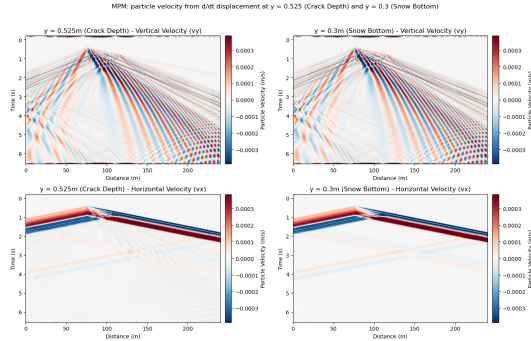


Figure: MPM-guided model velocity plots. Crack depth on left side, snow bottom on right. v_y at top, v_x at bottom. Velocity derived from displacement.

Simple model F-k plots and dispersion curves

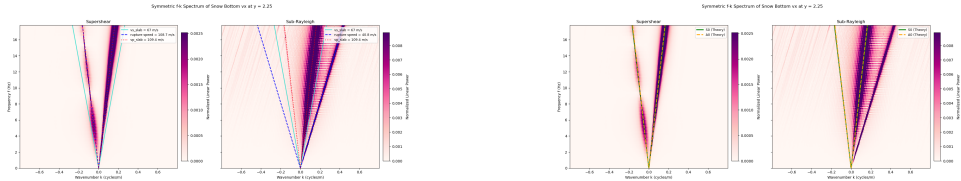


Figure: Supershear and Sub-Rayleigh F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

MPM F-k plots

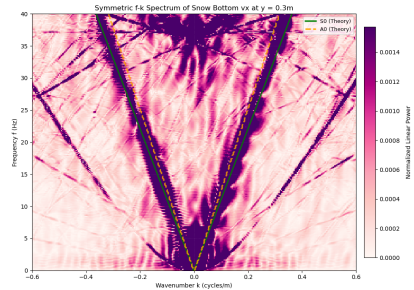
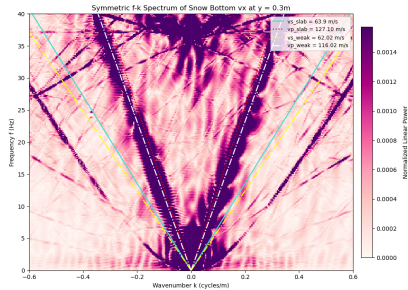


Figure: MPM F-k spectra with medium and Lamb-wave dispersion curves indicated by colored lines.

MPM STF cumulative Integral

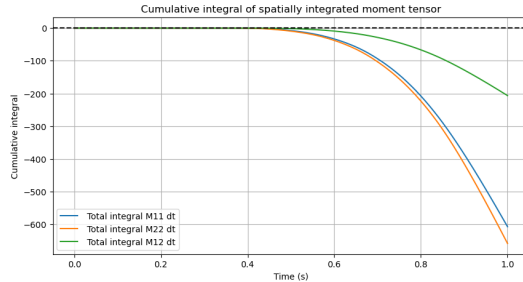


Figure: Cumulative integral over time of all MPM STF components



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